

Nainisha Bandari

Vagdevi Malineni

# Database Relationships & Schema Documentation

## 1. System Overview

The backend manages four core entities:

1. **Users** – Stores user account information.
2. **Products** – Stores clothing/product details.
3. **Measurements** – Stores body measurements used for AI fitting.
4. **Try-On History** – Stores AI-generated virtual try-on sessions.

These entities work together to provide a personalized AI-powered virtual try-on experience.

## Database Schemas

### • User Schema

#### Purpose

Stores authenticated user information.

#### Backend Responsibilities

- Registration
- Login
- JWT Authentication
- User Profile Management
- Measurement Ownership
- Try-On Ownership
- **APIs Using User Schema**

POST /api/auth/register

POST /api/auth/login

GET /api/users/profile

PUT /api/users/profile

## `</>` JSON

```
{
  "_id": "USR001",
  "name": "John Doe",
  "email": "john@gmail.com",
  "password": "hashedPassword",
  "gender": "Male"
}
```

- **Product Schema**

### **Purpose**

Stores all virtual try-on products.

### **Backend Responsibilities**

- Product CRUD
- Product Search
- Product Filtering
- AI Product Selection

### **APIs Using Product Schema**

POST /api/products

GET /api/products

GET /api/products/:id

PUT /api/products/:id

DELETE /api/products/:id

**</> JSON**

```
{
  "_id": "PRD001",
  "productName": "Blue Denim Jacket",
  "price": 2500,
  "availableSizes": ["S", "M", "L"]
}
```

- **Measurement Schema**

### **Purpose**

Stores body measurements for AI fitting.

### **Backend Responsibilities**

- Save measurements
- Update measurements
- Provide measurement data to AI Engine
- Size Recommendation

### **APIs Using Measurement Schema**

POST /api/measurements

GET /api/measurements

PUT /api/measurements/:id

DELETE /api/measurements/:id

**</> JSON**

```
{
  "_id": "MSR001",
  "userId": "USR001",
  "height": 170,
  "weight": 65,
  "waist": 32
}
```

- **Try-On History Schema**

### **Purpose**

Stores AI generated virtual try-on records.

### **Backend Responsibilities**

- Save AI Results
- Track User Activity
- Generate Analytics
- Maintain User History

**</> JSON**

```
{
  "_id": "TRY001",
  "userId": "USR001",
  "productId": "PRD001",
  "measurementId": "MSR001",
  "fitPrediction": "Perfect Fit"
}
```

## Schema Validation Rules:

### Users:

```
email: {  
  required: true,  
  unique: true  
}
```

### Products:

```
price: {  
  required: true,  
  min: 0  
}
```

### Measurements:

```
height: {  
  required: true,  
  min: 50,  
  max: 250  
}
```

### TryOnHistory:

```
userId: {  
  required: true  
}  
productId: {  
  required: true  
}
```

## Relationships:

### • User → Measurement

#### Relationship

1 User : 1 Measurement Profile

#### Mongoose Reference

```
userId:{  
  type:ObjectId,  
  ref:"User"  
}
```

#### Why?

Every user has a personal body measurement profile.

### • User → TryOnHistory

#### Relationship

1 User : N Try-On Records

#### Why?

A user can try multiple products.

#### Example:

User

└─ TryOn #1

└─ TryOn #2

└─ TryOn #3

## • Product → TryOnHistory

### Relationship

1 Product : N Try-On Records

### Why?

Multiple users can try the same product.

### Example:

Denim Jacket

└─ User A

└─ User B

└─ User C

## • Measurement → TryOnHistory

### Relationship

1 Measurement : N Try-On Sessions

### Why?

AI uses measurement data during every try-on session.

| Collection   | Purpose                   | Relationships                          |
|--------------|---------------------------|----------------------------------------|
| Users        | Authentication & Profiles | 1→1 Measurement, 1→N TryOnHistory      |
| Products     | Virtual Try-On Items      | 1→N TryOnHistory                       |
| Measurements | Body Dimensions           | N→1 User, 1→N TryOnHistory             |
| TryOnHistory | AI Session Storage        | N→1 User, N→1 Product, N→1 Measurement |

## **Architecture Benefits**

- Scalable MongoDB design
- Clean Mongoose references
- Supports AI workflows
- Easy populate queries
- Optimized indexing
- Supports future recommendation systems